The University of Azad Jammu and Kashmir, Muzaffarabad

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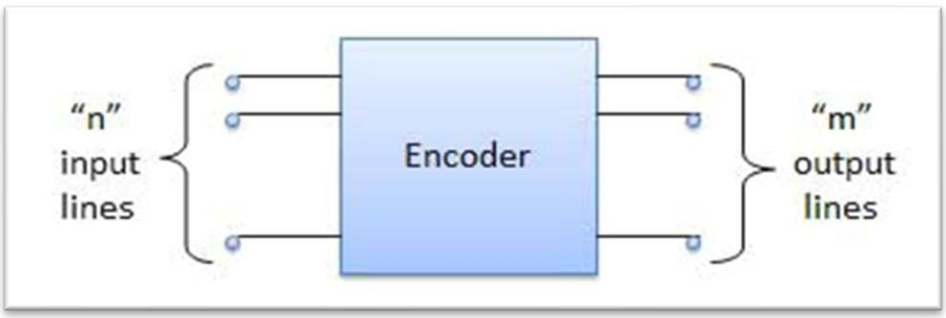
ROLL NO	**2024-SE-34**
COURSE TITLE	**Computer Architecture and Logic Design**
COURSE CODE	**CS-1205**
INSTRUCTOR	**Madam Sidra**
SESSION	**2024-2028**
LAB TITLE	**Encoder & Decoder**

Department of Software Engineering

Encoder: Encoder is a combinational circuit which is designed to perform the inverse operation of the decoder. An encoder has n number of input lines and m number of output lines. An encoder produces an m bit binary code corresponding to the digital input number. The encoder accepts an n input digital word and converts it into an m bit another digital word.

Block diagram



General Function:

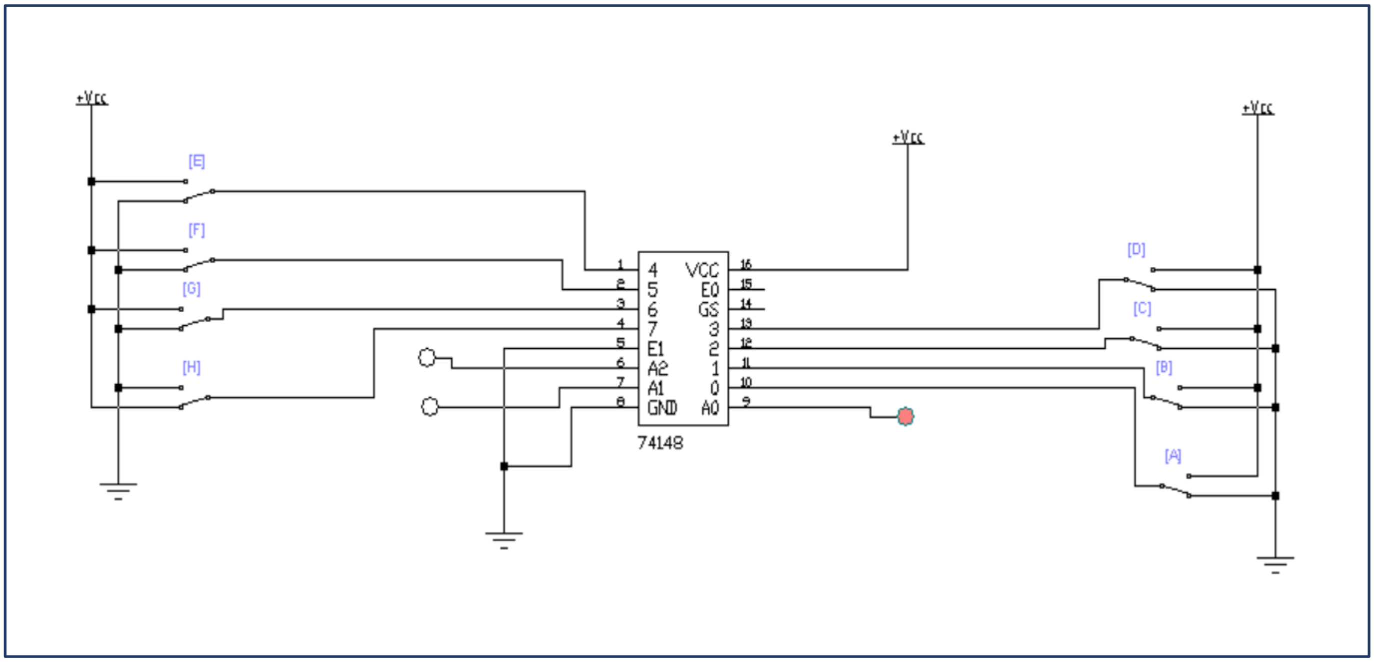
- Number of input lines = $\mathbf{2^n}$
- Number of output lines = **n**

Example: 8-to-3 Encoder

Truth Table:

D7 D6 D5 D4 D3 D2 D1 D0	C B A
0 0 0 0 0 0 0 **1**	0 0 0
0 0 0 0 0 0 **1** 0	0 0 1
0 0 0 0 0 **1** 0 0	0 1 0
0 0 0 0 **1** 0 0 0	0 1 1
0 0 0 **1** 0 0 0 0	1 0 0
0 0 **1** 0 0 0 0 0	1 0 1
0 **1** 0 0 0 0 0 0	1 1 0
1 0 0 0 0 0 0 0	1 1 1

Circuit Diagram By (EWB)



Output Expressions:

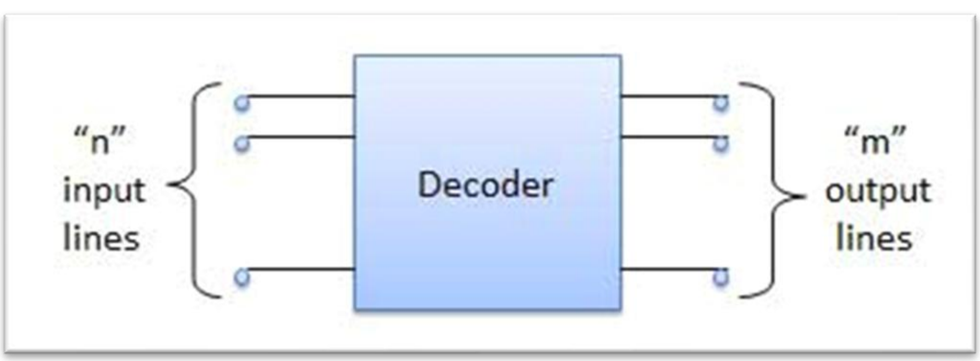
C =D4+D5+D6+D7

B =D2+D3+D6+D7

A =D1+D3+D5+D7

Decoder: A decoder is a combinational circuit. It has n input and to a maximum m = 2n outputs. Decoder is identical to a demultiplexer without any data input. It performs operations which are exactly opposite to those of an encoder.

Block diagram



General Function:

- Number of input lines = **n**
- Number of output lines = **2^n**

Example: 3-to-8 Decoder

Truth Table

C	*B*	*A*	*Y7 Y6 Y5 Y4 Y3 Y2 Y1 Y0*
0	*0*	*0*	*0 0 0 0 0 0 0 1*
0	*0*	*1*	*0 0 0 0 0 0 1 0*
0	*1*	*0*	*0 0 0 0 0 1 0 0*
0	*1*	*1*	*0 0 0 0 1 0 0 0*
1	*0*	*0*	*0 0 0 1 0 0 0 0*
1	*0*	*1*	*0 0 1 0 0 0 0 0*
1	*1*	*0*	*0 1 0 0 0 0 0 0*
1	*1*	*1*	*1 0 0 0 0 0 0 0*

Output Expressions:

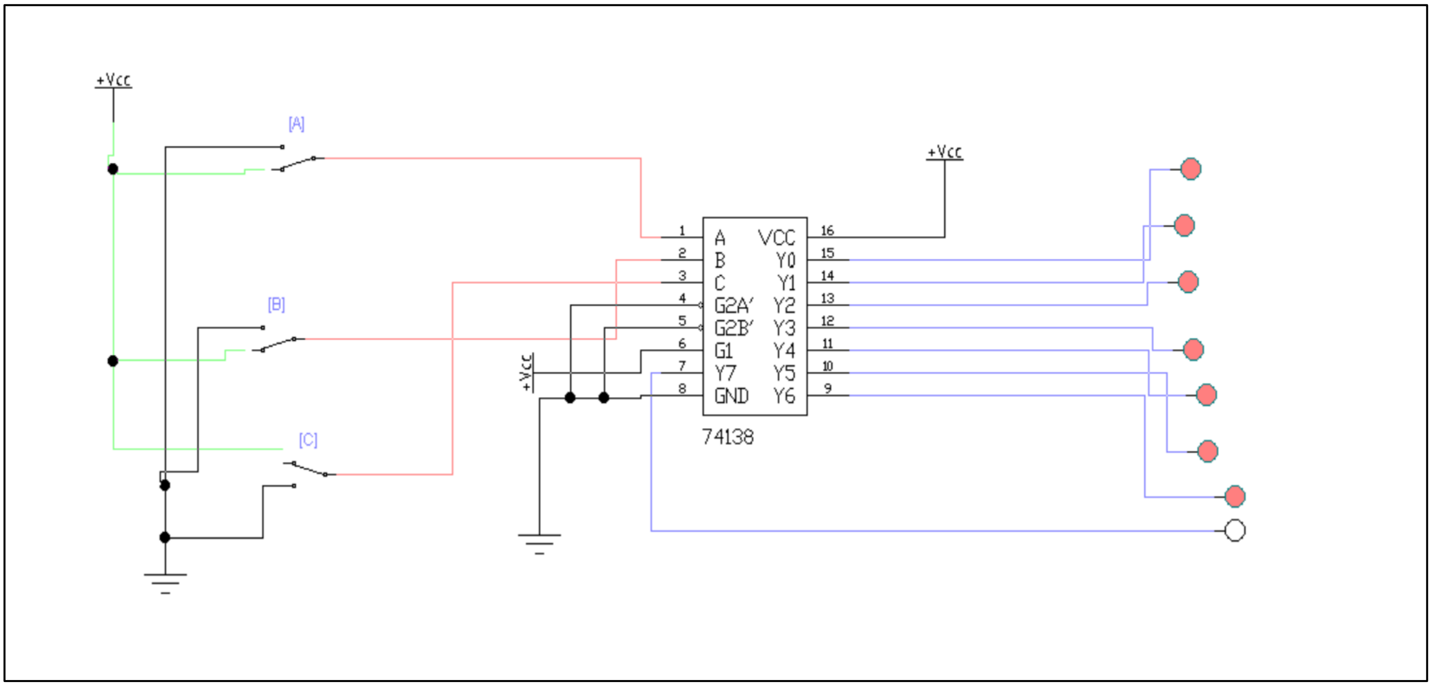
Y0=EB’A’

Y1=EB’A

Y2=EBA’

Y3=EBA

Circuit Diagram By using (EWB)



Best Of Luck